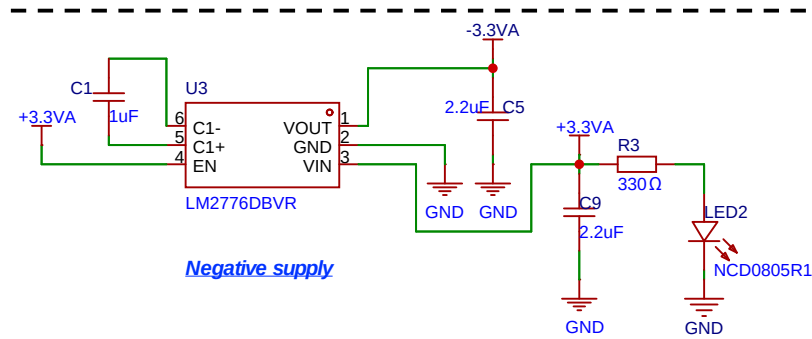
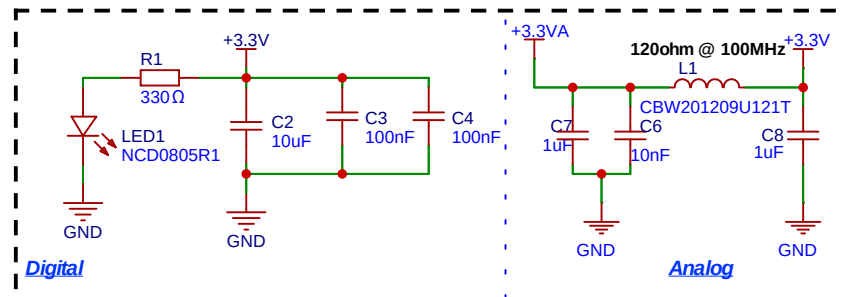
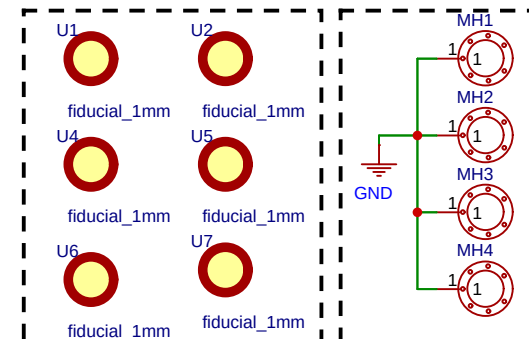
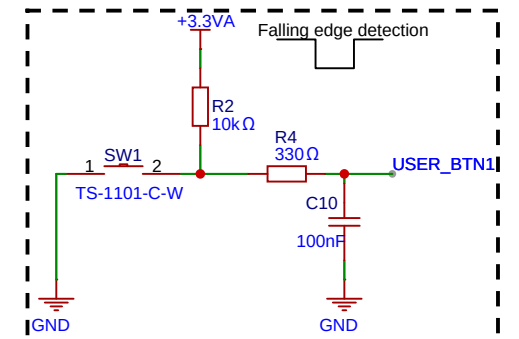




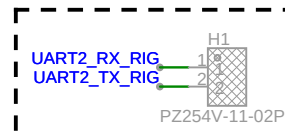
FRONT AND BACK FIDUCIALS MOUNTING HOLES



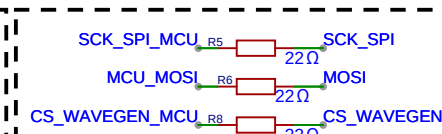
USER BUTTON



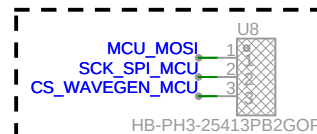
UART BREAKOUT PINS



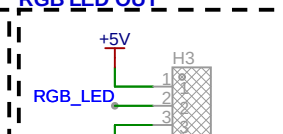
SPI TERMINATION RESISTORS



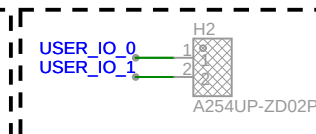
SPI TEST POINTS



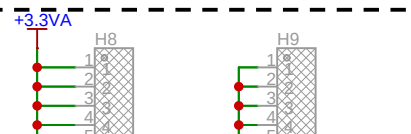
RGB LED OUT



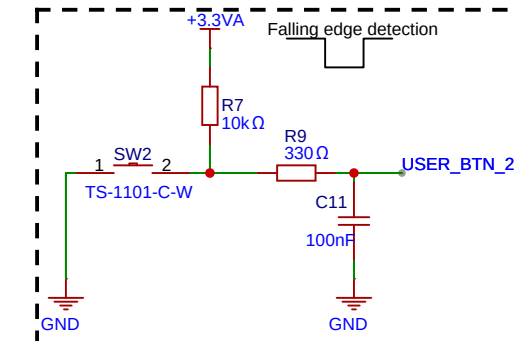
EXTRA USER IO



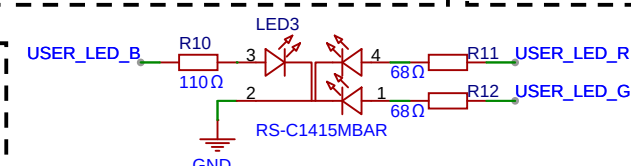
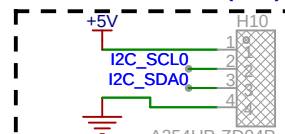
EXTRA POWER/GND RAILS



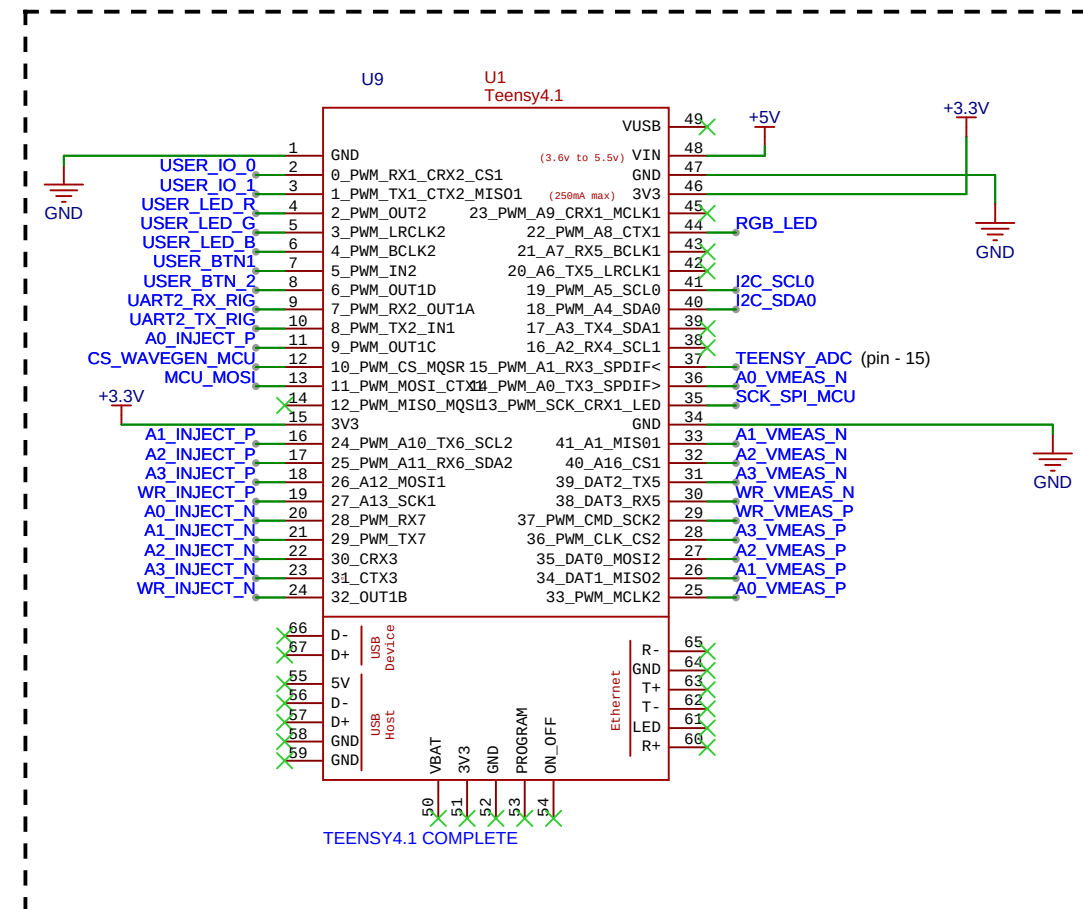
USER BUTTON



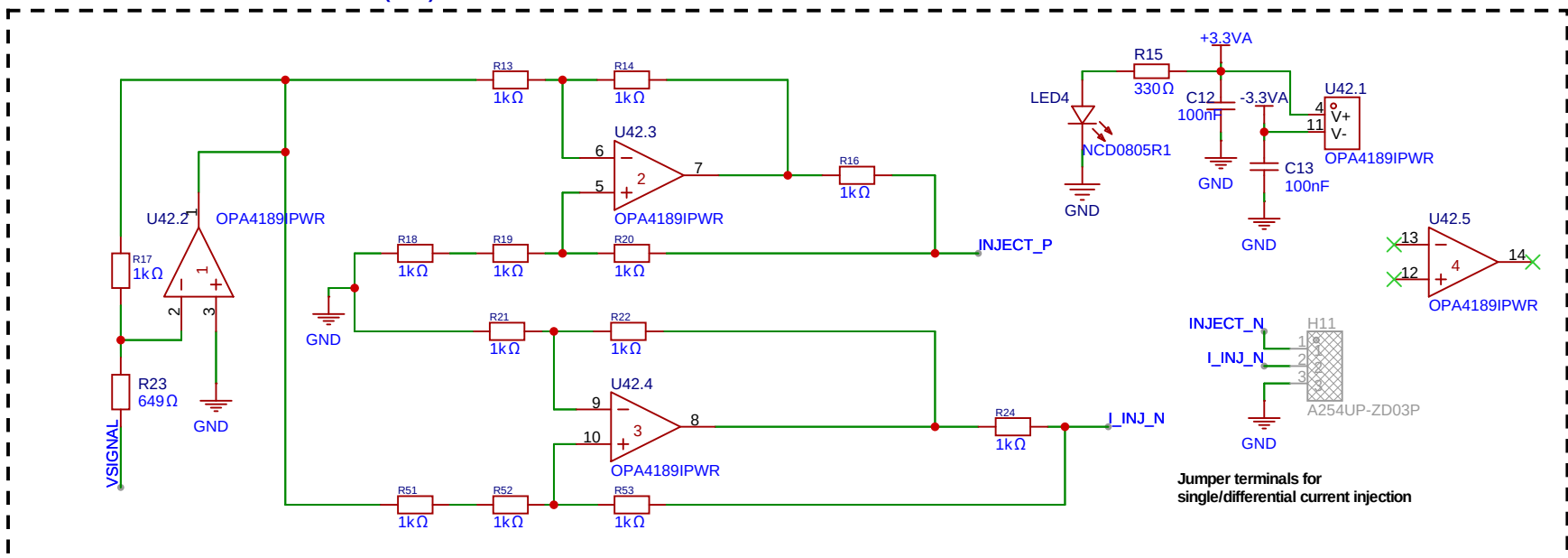
I2C BREAKOUT PINS (LCD)



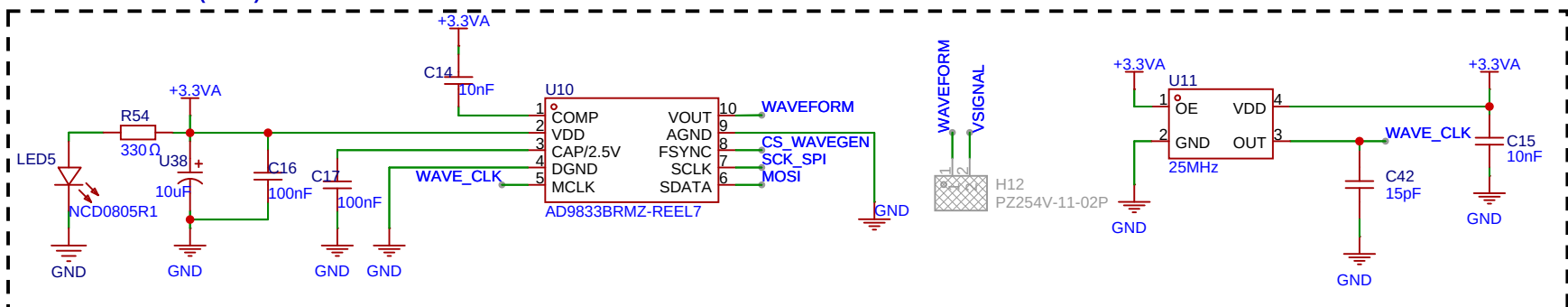
TEENSY 4.1 MCU



HOWLAND CONSTANT CURRENT SOURCE (1mA)



SIGNAL GENERATOR(0.65V)



NOTES

iEITBacon Too (pronounced "I Eat Bacon Too"; pun intended), is the second iteration of iEITBacon. The circuit is used to perform electrical impedance tomography (EIT)

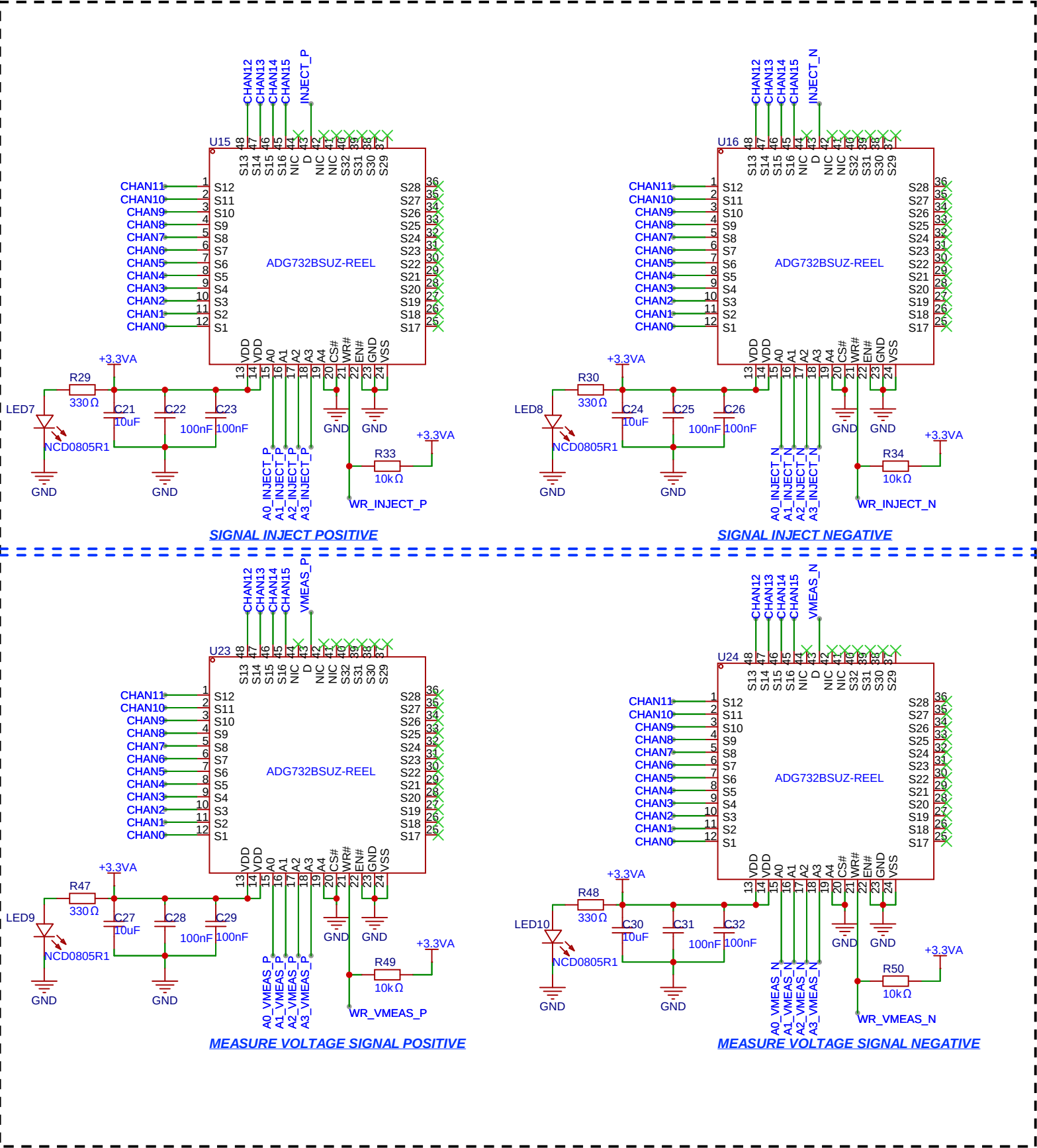
Tests conducted

- 1) GPIO channel control of muxes
- 2) SPI ports
- 3) RGB led ring
- 4) user led
- 5) user btn
- 6) Howland current source circuit using LM741
- 7) Vmeasure and filter using spice sim

Group	Change
Resistors	Power led resistor 220ohm to 330 ohm
Header pins	Added 5 extra user I/O
Header pins	Added extra power and ground rails
RGB	Changed to 5V

Schematic	iEITBacon Too v2			Create at	2025-10-29
				Update at	2025-11-10
Board	Board1			Page	MCU
Drawn		iEITBacon_Too			
Reviewed					
		Version	Size	Page 1 Total 3	
		V5.0		EasyEDA.com	

SIGNAL INJECTION AND MEASUREMENT MULTIPLEXERS



NOTES

MUX control

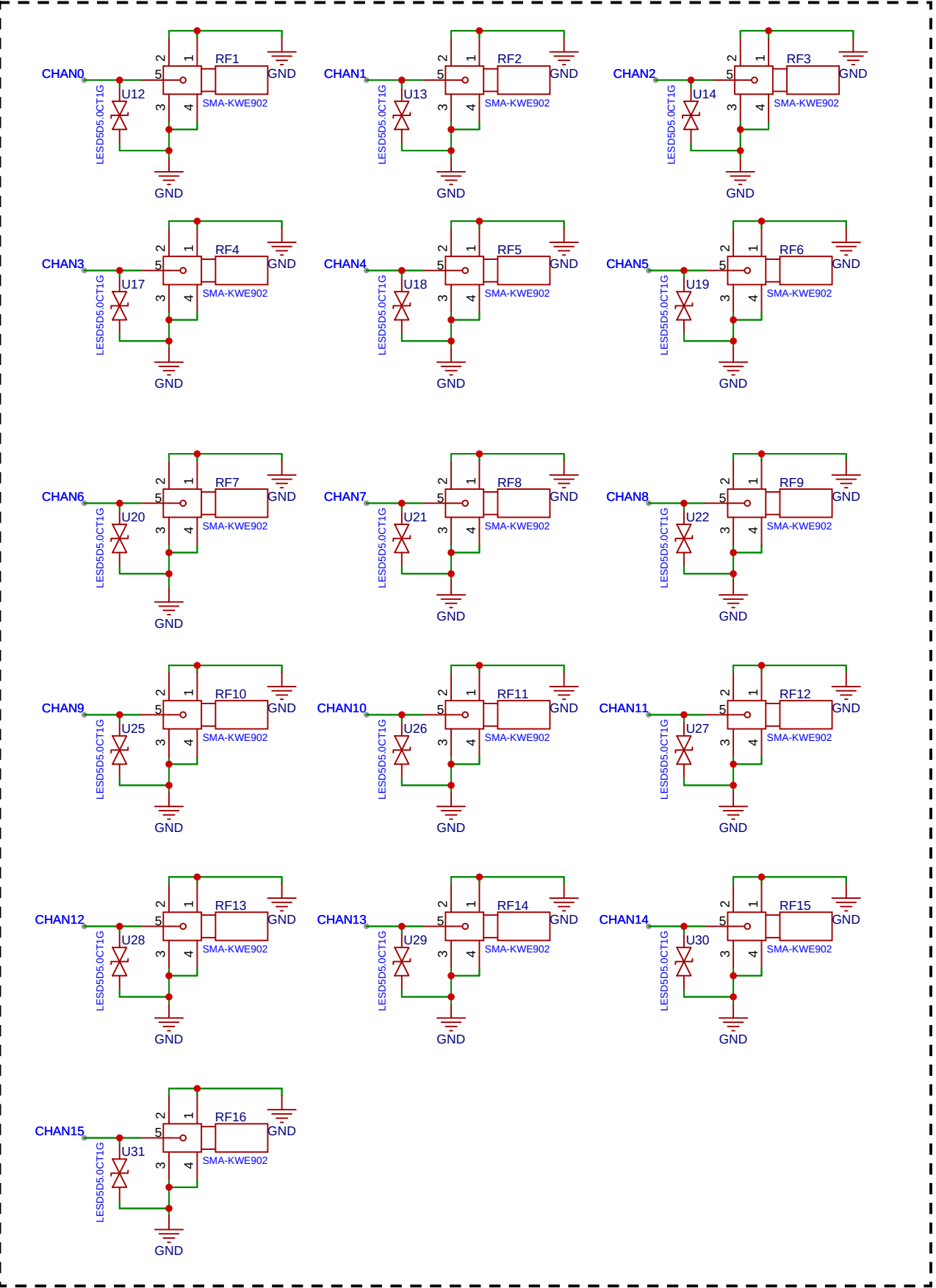
1) Pull WR pin low

2) Set logic control pins

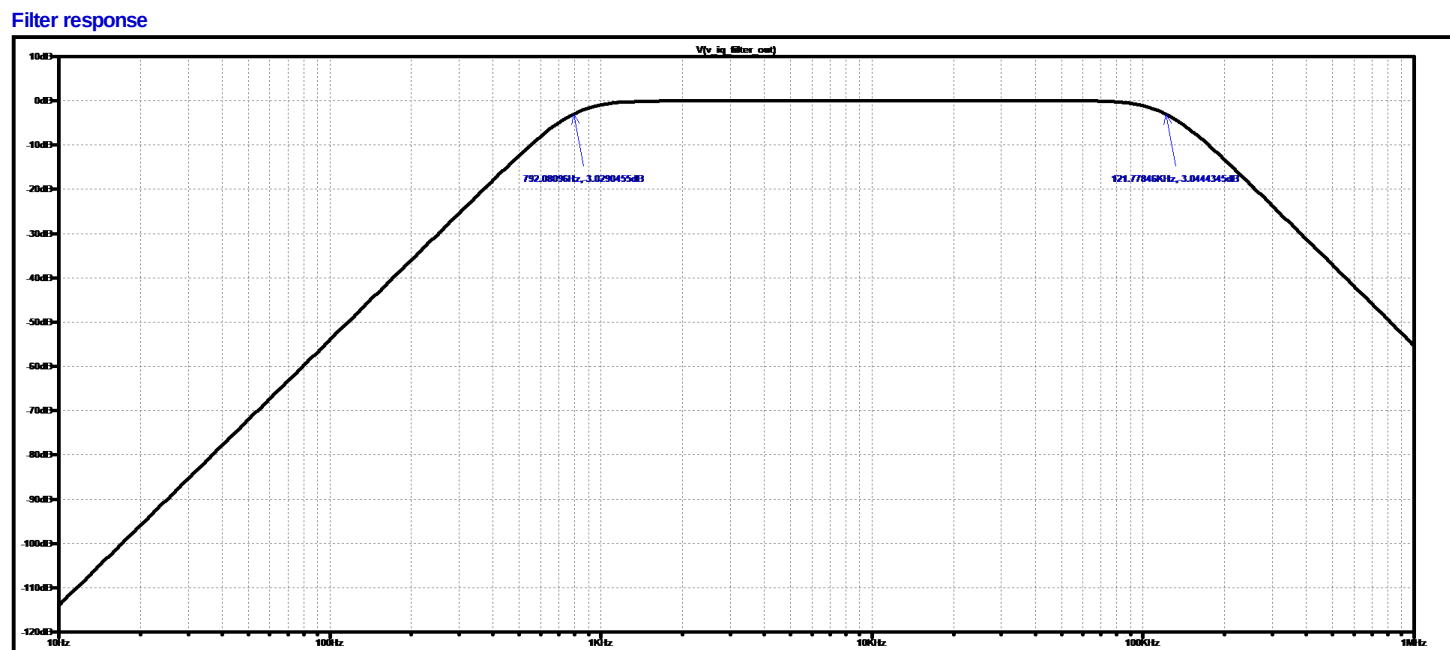
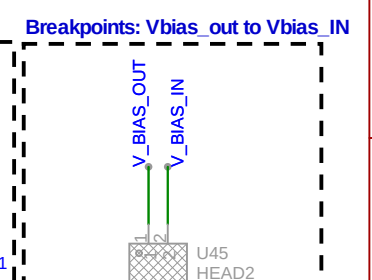
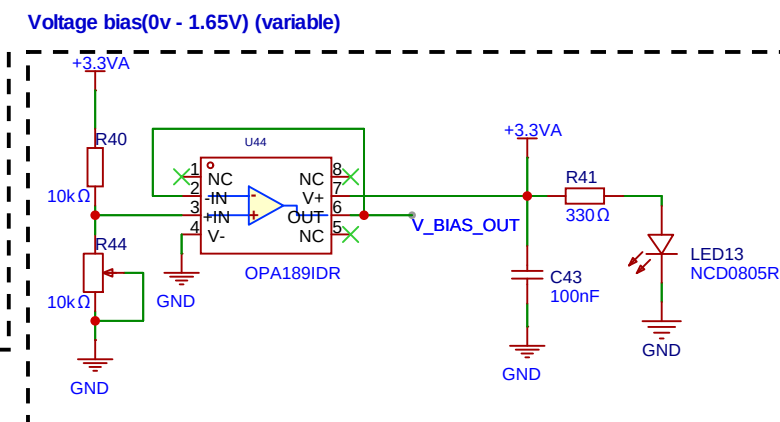
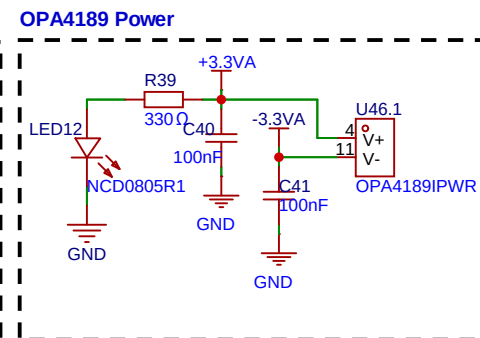
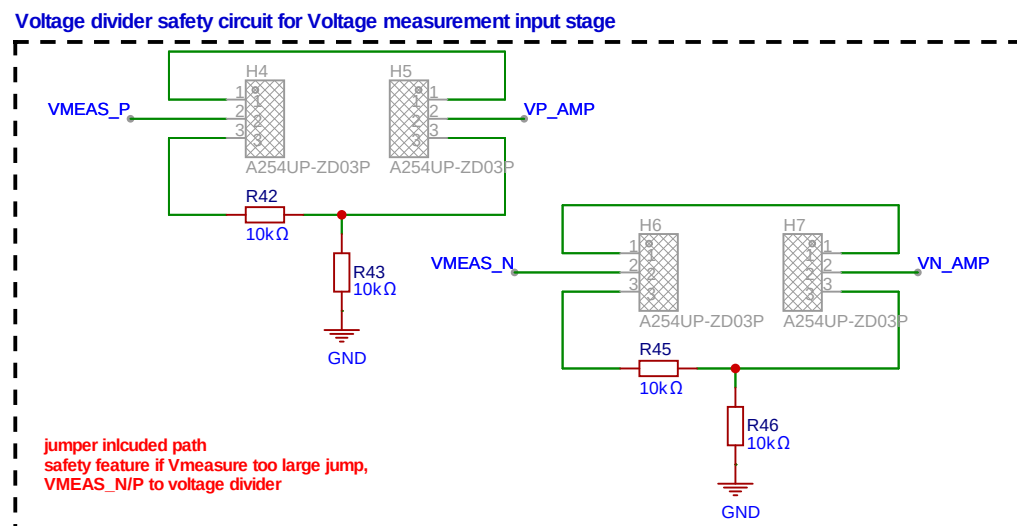
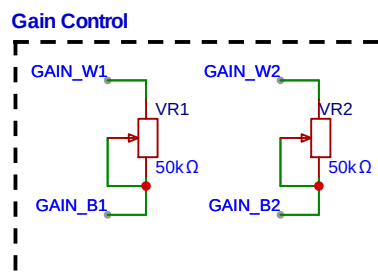
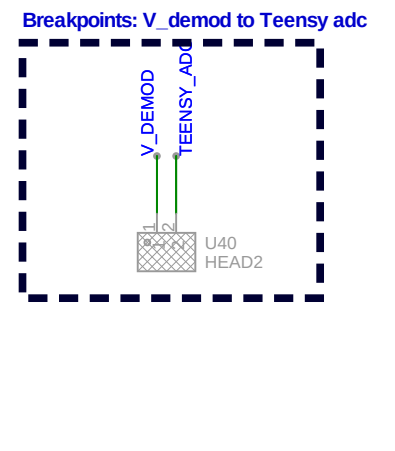
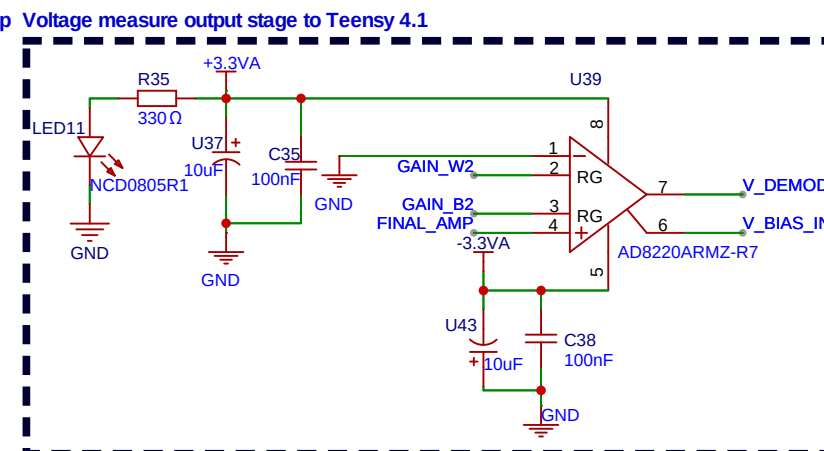
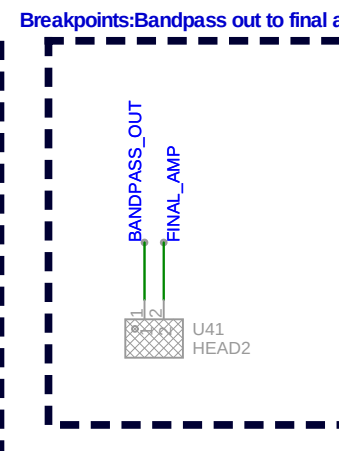
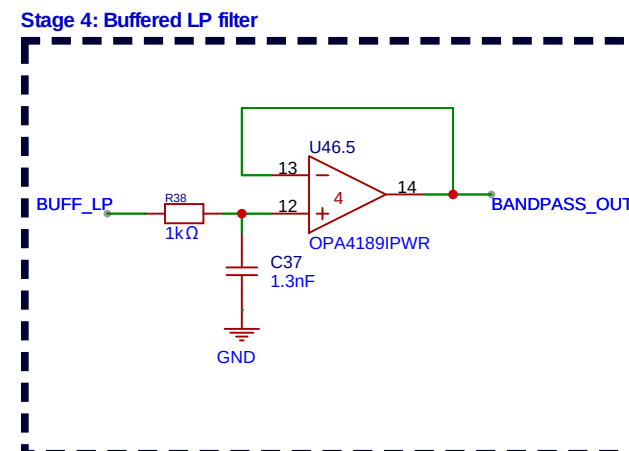
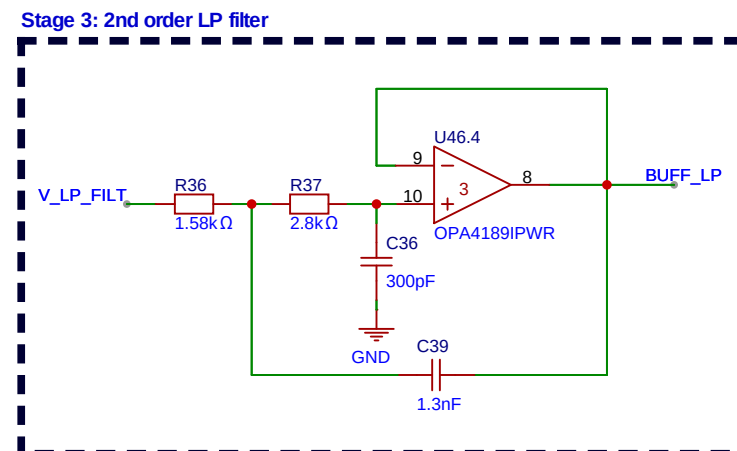
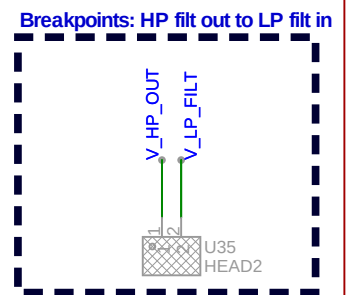
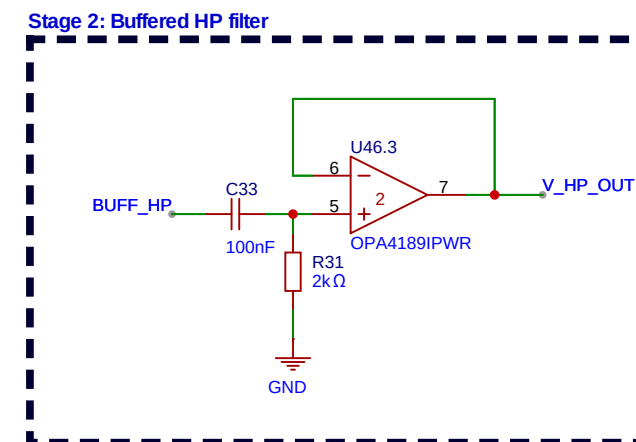
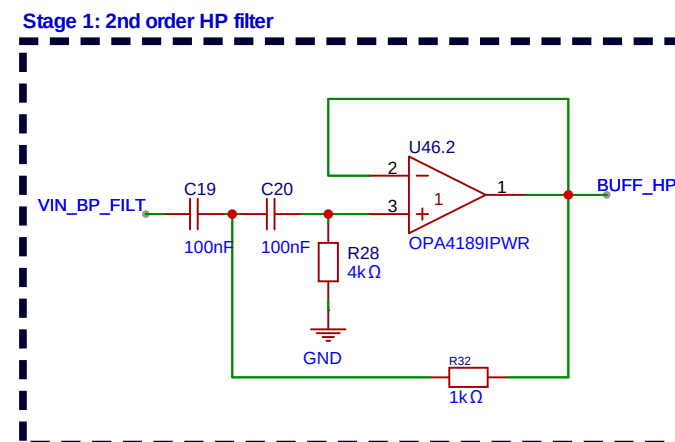
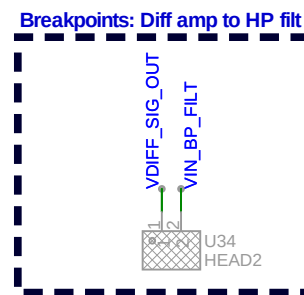
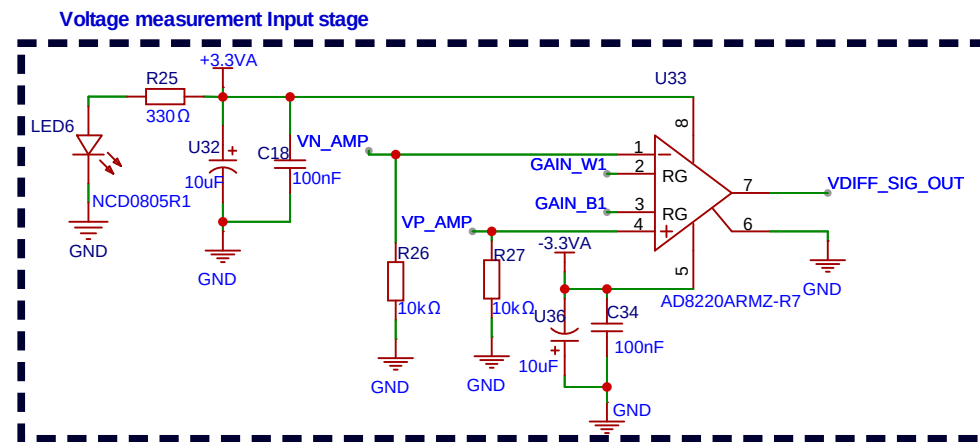
3) Pull WR pin high

Group	Change
Resistors	Power led resistor 220ohm to 330 ohm

SMA 90deg plugin (1/4"-36UNS-2A)



Schematic	iEITBacon Too v2	Create at	2025-10-30
		Update at	2025-11-10
Board	Board1	Page	MUXs
Drawn		iEITBacon_Too	
Reviewed			
		Version	Size
		V5.0	
		Page 2 Total 3	
		EasyEDA.com	



NOTES

Analog Devices → Analog Filter Wizard

Gain: 0dB
Passband: 120kHz @-3dB
Stopband: 1MHz @-40dB
CenterFreq: 9800Hz ($\sqrt{(800)(120\text{kHz})}$)
Fc_LP = 800Hz (1kHz with 20% safety margin)
Fc_HP = 120kHz (100kHz with 20% safety margin)
6th order Butterworth

Group	Change
Resistors	Power led resistor 220ohm to 330 ohm
Capacitors	Changed 10uF Diff amp to 10uF tantalum
Break points	renamed
V bias	opamp power led

Schematic	iEITBacon Too v2			Create at	2025-10-31
				Update at	2025-11-10
Board	Board1			Page	Voltage_measurement
Drawn		iEITBacon_Too			
Reviewed					
		Version	Size	Page 3 Total 3	
		V5.0		EasyEDA.com	